

HEROES, Design of an Exergame for Balance Recovery of Stroke Patients for a Home Environment

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Figure 1: Game concept of HEROES of the sea.

ABSTRACT

Stroke survivors suffer from balance impairment, causing an increased risk of falling. To recover balance, adequate stepping responses are key and these are practised by perturbing the patients during rehabilitation. Serious videogames, such as exergames, that train voluntary stepping can be found, but they don't allow to practice fast recovery steps in people with stroke. In this paper, we propose the design of a serious exergame (HEROES) to train stepping responses of stroke patients in a home environment. For this, we employed recent findings of action observation and motor imagery. We followed an iterative user-centred methodology to design the HEROES exergame. Stroke patients, physiotherapists and game designers were involved in every stage. The design of the HEROES exergame complies with the stroke accessibility guidelines, providing clear instructions and feedback. Therapeutic goals are defined by the progression of the level, ensuring to train paretic and non-paretic legs in a safe but challenging set-up.

CCS CONCEPTS

• **Human-centered computing** → *Empirical studies in HCI; Gestural input; User centered design.*

KEYWORDS

serious videogames, exergames, Stroke patients, rehabilitation

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1 RESEARCH DESCRIPTION

The HEROES project is a novel exergaming system for home use that allows people with stroke to increase their resistance to balance perturbations. The ultimate goals are to prevent falls and fall-related injuries, save fall-related health-care utilization and societal costs, and help people with stroke maintaining independence in daily life. HEROES will focus on improving stepping responses for recovering balance, capitalizing on recent findings that demonstrate the great potential of action observation and motor imagery. This approach sets HEROES clearly apart from any currently available home-based exergames, which aim at healthy persons and lack the required personalization.

Stroke survivors have different problems regarding motor skills, including falls with or without injury [6]. Daily life activities may be affected for a stroke survivor, such as walking, dressing, eating, and using the bathroom [4]. Because of these reasons, we can identify that there is a huge impact on the stroke population related to health care costs. Physical therapy is the principal form of rehabilitation for most people with stroke. The goal of physical therapy is to help patients relearn or improve simple motor activities such as walking, sitting, standing, lying down, and the process of switching from one type of movement to another. To accomplish this, physical therapists carry out training and exercises to restore movement, balance, and coordination [4]. Even though traditional therapies are necessary, literature indicates that playing video games during therapy sessions can increase the motivation to do therapy [2].

There is a specific type of video game for exercise, which are called exergames (exercise + videogames). Exergames have some advantages over traditional methods of exercising. They have different levels of difficulty and can focus on particular tasks of training, which offer variety to the patient. Depending on the performance of the participants, they can increase or decrease the task's difficulty. The possibility to personalize and choose gives the patient different challenges that are comfortable and suitable for them [10]. The use of commercial video games in rehabilitation has become popular in the last years. Examples are the Wii fit, Wii balance board, and the Xbox Kinect [1][3]. Nevertheless, commercially available exergames are not focused on the range of requirements of rehabilitation therapy, as they are designed for entertainment and fitness [9]. It is important to have additional tools for training stepping responses and improving balance as an extra method in rehabilitation. However, this would not aim to change or replace any traditional methods of exercising in rehabilitation [10].

To train balance in rehabilitation, stepping recovery is practised by perturbing the patients. Even though traditional therapies are necessary, literature indicates that playing video games during therapy sessions can increase the motivation to do therapy. Serious videogames, such as exergames, that train voluntary stepping can be found, but it remains a challenge to train recovery steps in a home environment. In this project, we proposed the design of a serious videogame to train stepping responses of stroke patients in a home environment, using recent findings of action observation and motor imagery. We followed an iterative user-centred methodology to design the HEROES exergame. It consists of 4 main stages, Empathize, Define, Ideate and Prototype. The design characteristics of the HEROES exergame emerged from the User-Centered design process, and are validated in every stage. During this process, stroke patients, physiotherapists, game designers and experts in human movement were involved. First, during the Empathize stage, we conducted a contextual study to understand the population and the problem to solve, interviews were conducted with physiotherapists and stroke patients. Then, using the contextual study results, we define the video game's logistical, technical, therapeutical and conceptual aspects. Next, we followed participatory design sessions where the stakeholders were involved to ideate game concepts of the therapeutic video game. Finally, we design a low-fidelity prototype producing an inexpensive version of the product. The design of the HEROES exergame complies with the stroke accessibility

guidelines, providing clear instructions and feedback. The therapeutic goal is achieved by the progression of the level design. The patient is able to train paretic and non-paretic legs in a safe set-up.

2 SPECIFIC RESEARCH OBJECTIVES

2.1 Objective 1: Understanding of the population

We conducted a semi-structured interview with 3 persons with stroke and 2 physical therapists to get their input for the user requirements in the serious game development. We transcribed and analysed the qualitative data using grounded theory techniques from the interviews and drafted a contextual study with the results. The main objective of engaging stakeholders is to design an exergame that has the potential to be effective in improving balance recovery and has the potential for later commercialization and implementation in general practice by fitting to the requirements of different identified stakeholders. People with stroke will be involved in identifying the user requirements for the exergame and during the subsequent design evaluations and iterations. They will also be involved in assessing the usability of the system and in its final evaluation. Physical therapists will be involved in the design of the exergame, in assessing the usability and they will guide the people with stroke during training with the exergame. Game developers and game designers will be consulted and collaborated with during the design of the exergame. Incorporating their knowledge and experience should result in an engaging, user friendly, and joyful game.

2.2 Objective 2: System requirements

During this research several studies have been done, all of them contribute to the design of the exergame system. We investigated the robustness of different sensory systems and developed an algorithm to measure stepping responses using the Real Sense¹ camera in comparison with a Motion Capture² system. This algorithm was evaluated in healthy subjects [8]. We investigated different ambience conditions that may affect the camera sensor. The goal was to evaluate what type of home-based ambience conditions could affect the skeleton tracker recognition [11]. We investigated different software that allows skeleton tracking with the camera sensor under various ambience conditions [8].

2.3 Objective 3: Design of the system and therapeutic goal

Several iterations of design sessions regarding the definition and the ideation of the game concept were held. We define the game design document for the development of the exergame. We held a brainstorming session to define the exercise for the therapy goal of the exergame. The objective was to get a clear description of the exercise, one or more scenarios of use, how the difficulty can be adjusted and what type of feedback should be provided. We validated the HEROES exergame concept. We conducted a validation session where 2 game concept ideas were presented. The goal was to get input regarding the likeability, playability and challenges of

¹www.intelrealsense.com

²www.qualisys.com

the game. We conducted a game mechanics brainstorming session with 1 physiotherapist, 1 stroke patient, 1 game designer, and 2 researchers to define the interaction that the player (patient) would have with the exergame. Once this was defined, we conducted several game-level sessions to define the progression of the exergame.

2.4 Objective 4: Engagement in-game and body ownership

We investigate different characteristics that are important to feel engagement in a 3D virtual avatar. Since the HEROES project aims to have a 3D virtual avatar it was of interest to evaluate the characteristics that a 3D virtual avatar should have for the user to feel engaged during the use of the system [12]. We investigated the effect of a personalization system to configure automatically the user avatar. The aim is to compare the automatic system against the manual personalization of the avatar for game experience [7].

3 METHODS

For the design of the exergame we followed a User-Centered methodology inspired by Design Thinking (as shown in Figure 2), which consists of 4 stages: Empathize, Define, Ideate and Prototype. As part of the empathize stage, semi-structured interviews were designed following the PACT framework [5] and conducted with physiotherapists and patients. The interviews were analyzed using qualitative techniques such as affinity diagrams, which provided information for the define stage and the data was prioritised with the MoSCoW method. During the ideate stage, we set the boundaries regarding the exercise definition and conducted a design session with the objective of establishing the first design proposal for the exergame. Finally, in the prototype stage, a low fidelity prototype is ready for validation.

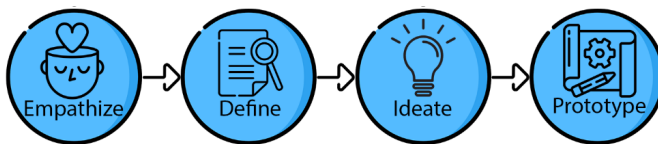


Figure 2: User-Centered methodology for the design of therapeutic exergames.

3.1 Empathize

The main focus of this stage was to understand the context of the patients as end-users. There is a need to understand the problem to solve. This was done with a literature review, interviews with specialists and stakeholders, and through observation sessions of physical therapy. We gathered all the necessary information to get an understanding of the population. As a result of this phase, we got a contextual study, which is an analysis that helps us characterize the problem to solve.

3.2 Define

For the home-based therapeutic exergame, we proposed to define 4 important topics to have into consideration. First, is the Logistical, which included the definition of the team and ethical and/or medical

committee. Second, the Technical, we specified the technical part of the system, having in mind all desirable features. Following, the Therapeutic, it is crucial to define considerations regarding the therapeutic objective to achieve a safe therapeutic exergame. Finally the Conceptual, here we described the conceptual aspects, such as items, objects, and characters, in the game. It should be documented not only what they do, but also what they affect, how they interact, and their role and behaviour in the game.

3.3 Ideate

Once the videogame is defined, we looked for alternative ways to view the problem and identify innovative solutions to the problem statement. To achieve this, an approach to follow is the participatory design and the therapeutic design. Participatory design sessions attempt to actively involve all stakeholders (e.g. patients, therapists, experts, etc) in the design process. This helps to ensure that the result meets their needs and is usable. The other aspect is to fulfil the therapeutic goal. To ideate and design something suitable for the patient, we involved specialists in the topic, such as therapists or physiotherapists. In this process, we used Virtualization, which is the implementation of a virtual exercise by defining input (tracking) and output (feedback) requirements [9].

3.4 Prototype

In this phase, we produced some inexpensive, scaled-down versions of the product and a user manual with the playability of the videogame. It is important to remember that this methodology works in iterations. At this point, we can return to the ideate process and make the corresponding modifications to continue with the prototype.

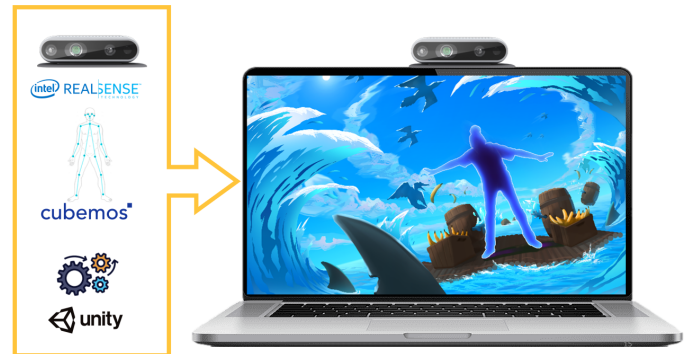


Figure 3: Prototype of the HEROES exergame with the technical requirements

4 RESULTS AND CONTRIBUTIONS TO DATE

There has been several outcomes from the different stages of the project, all leading to the design of the exergame.

A contextual study was conducted, in which semi-structure interviews to patients and physiotherapists were held. Some of the design characteristics that emerged from the interviews were focused on the configuration of the game. The system should allow to personalise the difficulty level, which includes the perturbation

type, perturbation characteristics and the performance (i.e., the quality of the step), the session time and the avatar customization. Because of the age of the participants, a more realistic game concept is key. Visual stimuli are very important and elements that may look dangerous (e.g. tornados, sharks) should be avoided. To keep the patient motivated to play the game, feedback should be provided and patients should be able to see their progression. Regarding the therapy goal, reactive stepping should be accomplished by mimicking a virtual avatar in the game following the principles of action observation and motor imagery. The most important characteristics to keep in consideration are:

- The system should have configuration options, such as, enlarge letters, text to speech, dim brightens, tutorial session and FAQ.
- The instructions should be short and clear.
- The scoring system should be simple, indicating if the movement was Good, Normal or Bad.
- To ensure that the therapeutic goal is accomplished the system should provide the recording of session to be able to replay and improve.
- The system should be safe and easy to use. The device and set up should be as small as possible, providing a hands free scenario and Plug and play settings.

During the co-Design of the game, together with physiotherapists and patients, experts in video game design and human movement experts, we defined the therapeutic exercise characteristics (see Figure 4).

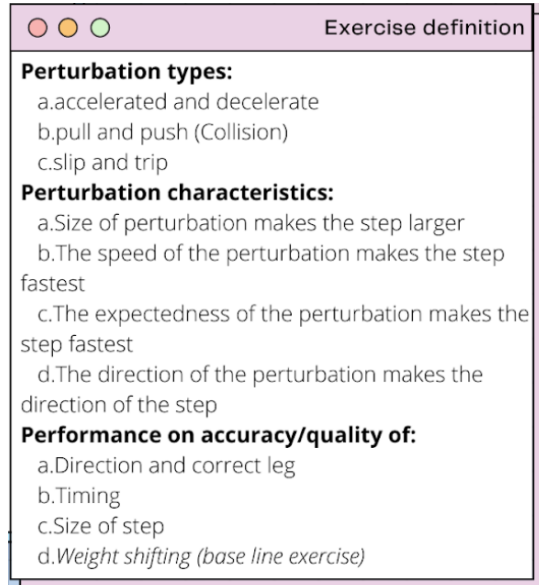


Figure 4: Exercise definition for the design of the exergame

A first version of the Prototype is ready for evaluation with the technical requirements that were defined in previous stages. At this point the exergame system is implemented to use a Depth Camera (RealSense), with a Skeleton tracker software (Cubemos), and develop in the game engine Unity (See Figure 3).

5 NEXT STEPS

The expected next steps goes towards the evaluation of the exergame, how to evaluate for improvement and getting a better version of the system.

We are planning on evaluating the system in 4 stages, First we will evaluate in healthy young subjects, then we will evaluate in healthy old subjects, Until here we will ensure that the videogame is safe to try for the patient population.

After the safety study is done, we will evaluate the usability and game experience with patients and therapists in close collaboration with research clinics in the Netherlands in the upcoming years.

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